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Description - Contains theory for the surface coils. Sheet numbers listed at the end of some subsection titles refer to the sheet number within the TPS/RF block diagrams.

1- SURFACE COIL OPERATION

The linear surface coil differs from the quadrature coil due to the fact that it has a static RF field versus a rotating one. In the linear surface coil design the RF input would be placed across one of the capacitors on the end rings (see Illustration L1946A).

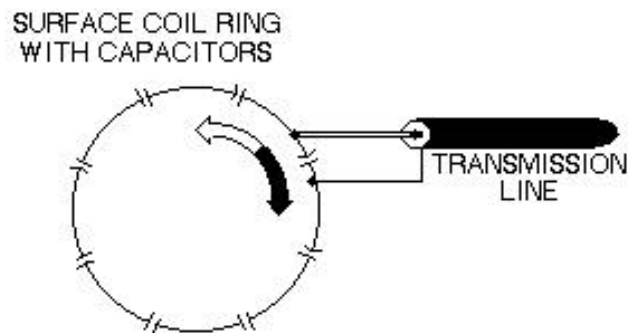


ILLUSTRATION L1946A
LINEAR SURFACE COIL RF INPUT SIGNAL

Surface coils, closely coupled coils, hear both the signal and the noise over a smaller area. The use of surface coils increases the signal-to-noise ratio (SNR) of images. This occurs for two reasons: (1) the coil is closer to signal-emitting protons and, therefore, receives a strong signal; (2) the coil hears only noise originating near the coil, which is less than the noise that is picked up from the entire body by the body coil. There are, however, two drawbacks to the use of these smaller coils. One is that the usable FOV (field of view) of the image is limited to the field of sensitivity of the coil (this is also what limits the noise coupling). In other words, if the coil picks up signal/noise only from a 12-centimeter long area, then there is no sense in using it to image anatomy that is 48 centimeters long. A surface coil with twice the FOV could be used and the number of acquisitions and positions could be cut in half. There would still be a time penalty, as well as a reduction in SNR, because the larger coil would "hear" more noise.

The other drawback holds true for an FOV in the depth (A/P) direction. If the coil receives signal/noise to a depth of only twelve centimeters, it does not make sense to attempt to image anatomy that is beyond this. The trade-offs are similar with regard to increasing coil size (increased size increases the depth of the FOV); however, the major difference is that the SNR is on a gradient with respect to distance into anatomy from the coil. This type of nonlinearity of signal can be very troublesome when you are trying to image volumetric types of anatomy (e.g., the pelvis).

The surface flex coil consists of two 5" x 6.5" linear surface coils that are electronically summed to create a co-rotating saddle coil pair (curved loops where the currents in both halves circulate in the same direction). These are placed in a soft, flexible package that can be wrapped around the anatomy of interest.

2- MULTICOIL ARRAY OPERATION

The initial release of the Phased Array option consists of a group of coils and receivers whose individual images are combined to create one image with improved SNR and increased FOV capability. The coils are configured in either a longitudinal fashion, to stretch the length of the useable FOV of the coil, or in a volumetric fashion, to improve the uniformity of signal across a volume.

When these coils are used for imaging, the anatomy selected for the scan is excited with the body coil, as in any other exam; however, each small individual coil receives the signal/noise only from its own useable FOV. The outputs of the individual coils are simultaneously transferred to each respective receiver and are separately processed. These four different output images are then combined and a single image is created.

When you are using a phased array for increased length of FOV, the normal acquisition matrices may not provide small enough pixel sizes. This consists of 512x512, 512x384 and 512x256 (Rectangular FOV compatible with 512x512). In addition to the phased arrays, you will also need three extra receivers (in addition to the one already in your Signa system), and a total of 128 megabytes of TPS memory. There are also several hardware modules in the system to amplify and switch the receiving signal before returning to the receivers. See Illustration L1947A.

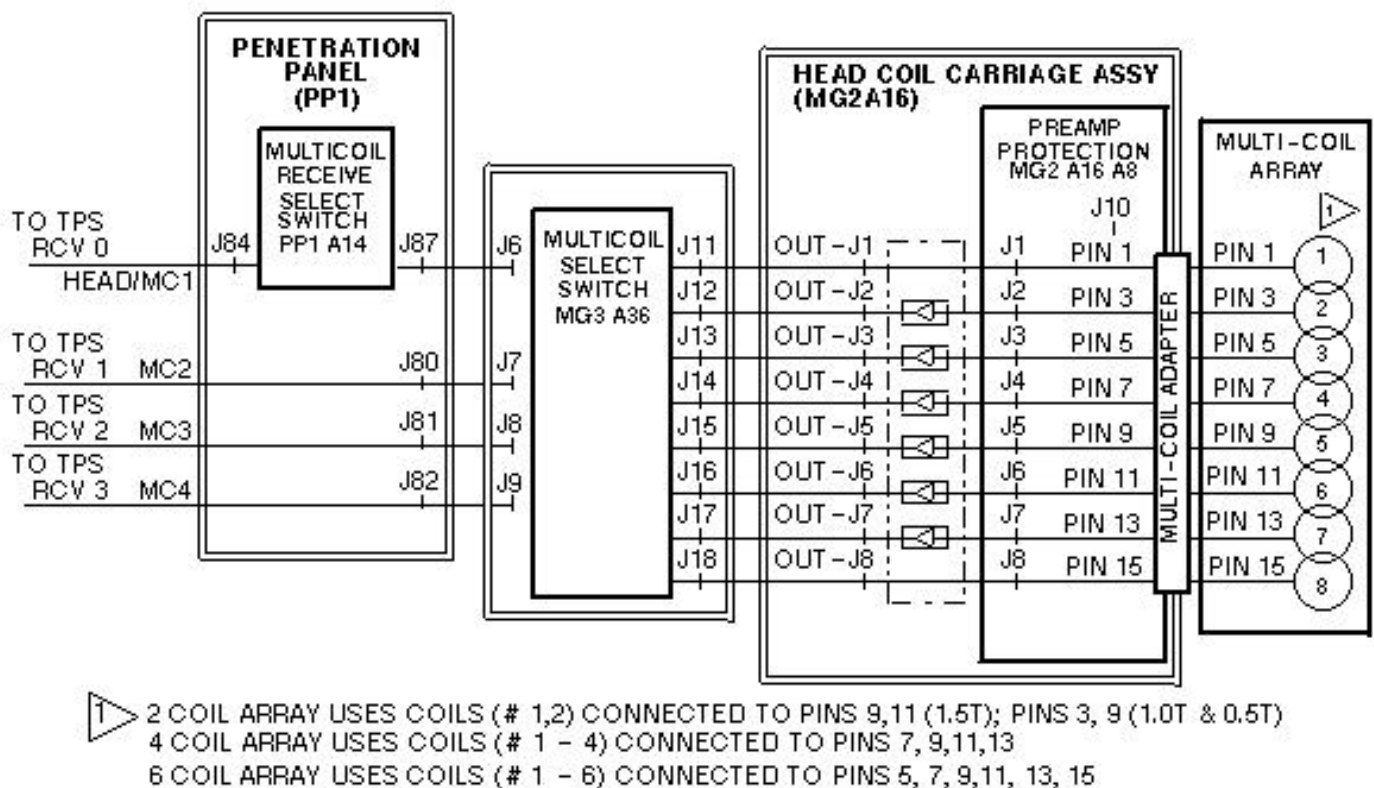


ILLUSTRATION L1947A

MULTICOIL ARRAY TO RECEIVER CONFIGURATION

The multicoil arrays are made up of an input blocking circuit, the antenna loop inductance and loss and the distributed capacitance. There is also a 50-ohm phase shift network which ensures a multiple of a half wavelength from the 30-pin connector to the coil input and an isolation balun (bridge balun) that isolates the cable system ground from the coil ground.

The geometry of the coil is chosen to optimize performance. This geometry dictates the loop inductance and loaded coil losses. The distributed capacitors are chosen to resonate with the loop inductance. The input capacitor of the coil (C_{in}) is chosen such that the loaded coil looks like 50 ohms at the preamp input. The air wound blocking inductor is dictated by the value of C_{in} such that the blocking circuit resonates at 63.86 MHz (1.5T), 42.68 MHz (1.0T), or 21.29 MHz (0.5T).

The input circuitry for each antenna element in the array serves two functions. The first is to match the input impedance of the individual antenna loop elements to the desired value of 50W (see Illustration L1948A). This is accomplished through the use of a simple ell matching network consisting of a series inductor L_{in} and shunt capacitor C_{in} . The second function performed by the input circuitry on the array is that of transmit blocking. This is actually performed by both the input matching network and the transmit blocking network, thereby providing twice the current blocking of a single network at a given bias level.

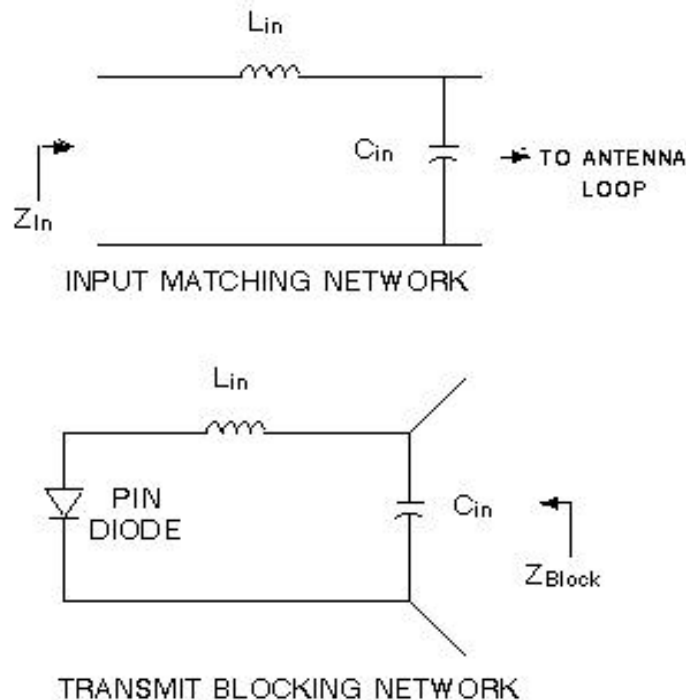


ILLUSTRATION L1948A
INPUT CIRCUITRY NETWORKS

The need for a transmit blocking network arises from basic electromagnetic theory. The transmit magnetic field excites currents in the individual antenna loop elements of the array. These loop currents will then re-radiate their own small magnetic fields, which constructively and destructively add to the existing transmit signal. This interference causes an undesirable inhomogeneous transmit magnetic field. Minimization of the undesired magnetic fields is accomplished through the minimization of the excited antenna loop currents. This is accomplished by switching high impedance networks into the antenna loop during transmit. These are the transmit blocking networks.

The transmit blocking network (and input matching network) are simple high impedance parallel resonant LC circuits tuned to resonate at $f_o=63.86$ MHz (1.5T), 42.68 MHz (1.0T), or 21.29 MHz (0.5T). The resonant circuit is closed (or switched into the antenna loop) during transmit by forward biasing the PIN diode. This bias current is supplied from the multicoil driver board and is split equally between the two diodes (one for each network).

During receive mode operation, it is important to the overall performance of the multicoil array to decouple the individual antenna loop elements. This is accomplished through two different methods. For adjacent antenna loop elements, the decoupling is accomplished geometrically. The magnetic flux linking two adjacent loops is canceled by adjusting the overlap spacing between them. For both adjacent and nonadjacent antenna elements, a second method of decoupling is used.

The low noise preamplifier (46-264442) used in the Signa MRI system has a very small real input impedance (less than 6W). If the electrical length of the system between the preamp input and the input matching network of the antenna loop is kept at a multiple of a half-wavelength, then the impedance seen at the diode terminals of the antenna loop's input matching network will also be small and real. This small real impedance at the diode terminals combined with L_{in} and C_{in} of the input matching network form a "lossy" parallel LC blocking network. This receive mode blocking impedance reduces the individual antenna loop current which in turn reduces the magnetic flux produced by the loop. This reduction in magnetic flux in turn reduces the coupling between antenna elements. Although this decoupling method is not as good as the geometric decoupling used for adjacent loops, it does provide loop decoupling and is needed for overall array performance.

3- MULTICOIL SWITCH DRIVER BOARD – SHEET 7

The multicoil switch driver (MCD) board provides a switched DC bias signal that controls the T/R network associated with the coil arrays. The drivers enable and disable pin diodes on the multicoil arrays to protect the preamps during transmit of RF pulses or enable the preamps to receive the signals. The board is located in the dynamic disable chassis in the RF cabinet.

The multicoil switch driver board supplies +5 volts on the output (J3) when either unblank signal is high. During receive mode of the multicoil array, when both unblank signals are low and the drives are enabled, -5 volts is supplied on the output. There is error detection circuitry for short and open circuits on each drive. The error detection asserts the signal OP1FAIL* if error reporting is enabled for that drive. The OP1FAIL* signal corresponds to bit 2 of register 4 (FAULT) on the dynamic disable/TR driver board (46-264808). The drives may be enabled to supply +5 volts and -5 volts controlled by the unblank signals, or it may be disabled to supply +5 volts constant. Each drive may be set in one of these four states:

Enabled, Error Reporting enabled— normal multicoil in use
Enabled, Error Reporting disabled—use for experiment coil
Disabled, Error Reporting enabled—use for more than four coils
Disabled, Error Reporting disabled—multicoil not selected

The receive select output (J4) controls a relay switch at the penetration wall which selects between the head coil preamp and the multicoil preamp, which share receiver 1. Whenever any one of the four drives is enabled, the receive select line should supply +5 volts to select the multicoil preamp. Otherwise, the receive select output should be 0 volts and the head coil preamp is selected.

Bidirectional communication is done with the dynamic disable/TR driver board through connector J1. The dynamic disable/TR driver board communicates with the IPG through the SPC board via the fiber optic (MDS) link with differential drivers/receivers. Most of the signals from the fiber optic (MDS) link are passed to the multicoil driver board via single-ended TTL signals.

The port enable register is a read/write control register selected by address chip select SPDEC3. The register is accessed at address 09H relative to the dynamic disable/TR driver board base address. The power on default is 00H, ports disabled and error reporting enabled.

The load fault register is a read only status register selected by address chip select SPDEC4. The register is accessed at address 0AH relative to the dynamic disable/TR driver board base address. These bits are active low. The power on default is 0FFH, no errors.

The driver outputs have error detection for short and open circuit faults. The voltage drop across a 10-ohm resistor is used to determine the amount of current that is drawn. The reference resistors are set such that an open error will occur for current less than 50 mA and a short error will occur for a current greater than 700 mA. The error detection is bidirectional; it works for both +5 volts and -5 volts output of the drives. The results of the error detection go through boolean logic, and are clocked into the register outputs of a PAL on a transition edge of the unblank signal. The output of the PAL is the load fault register. The register is reset by OP-RST* or OP-FRST. The clock (OCLK) used by the PALs is an exclusive OR of the unblank signals and negated OP-DBLK signal. New values are clocked into the PAL register on the falling edge of OCLK.

4- MULTICOIL RECEIVE SELECT SWITCH – SHEET 9C

The multicoil receive select switch connects one of two RF receive lines (either head receive, or multicoil receive #1) to the first TARDIS receiver in a multicoil system. Inside this assembly is a single-pole, double-throw relay, operated by a +5 volt signal (provided by multicoil switch driver board (refer to Section 3- Multicoil Switch Driver Board)).

The multicoil receive select switch mounts to the system cabinet I/F panel, MR2A11, for all mobile systems because the mechanical relay is magnetic field sensitive at the penetration panel, PP1.

5- MULTICOIL PREAMP SELECT SWITCH BOARD – SHEET 9C

The multicoil preamp select switch board was designed to be used with the multicoil compatibility kit. It resides at the base of the magnet pedestal.

The board performs three functions:

It selects four out of eight preamps to be connected to the four multicoil receivers. This is done using four T-configuration PIN diode switches. Considering SWITCH1, if the signal $IN1-IN2^* = +3.4V$, then CR12, CR26, and CR28 are on (RF short) and CR11, CR25, and CR27 are off. This means that signals from J11 to J6 are allowed to pass, and signals from J15 to J6 are blocked. The opposite is true when $IN1-IN2^* = -4.0V$.

The output of the unselected preamp is terminated with 50 ohms via R18 or R19.

Power (+15V) is supplied to the preamps via L23 and L24.

6- MULTICOIL PREAMP PROTECTION BOARD– SHEET 9C

The multicoil preamp protection board was designed to be used with the multicoil compatibility kit. It resides in the head coil carriage assembly. It protects the preamps from high RF voltages developed on the coil during transmit and it provides the multicoil array with T/R bias.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 27, 1998	M. Whitlow	Initial conversion from Toolbook to Word.